Software Architect

Understand the Basics

- What is Software Architecture
- What is a Software Architect
- Levels of Architecture
  - Application Architecture
  - Solution Architecture
  - Enterprise Architecture

Software Architecture

Describes how an application is built including its components, how they interact with each other, environment in which they operate and so on.

Software Architect

A software expert who makes high level software design choices, including technical standards, tools, design principles, platforms to be used, etc.

Responsibilities

- Tech Decisions
- Design & Architecture Decisions
- Requirements Elicitation
- Documentation
- Enforcing Standards
- Collaborate with Others
- Consult & Coach Developers

Important Skills to Learn

- Design & Architecture
- Decision Making
- Simplifying Things
- How to Code
- Documentation
- Communication
- Estimate and Evaluate
- Balance
- Consult & Coach
- Marketing Skills

Technical Skills

Programming Languages

- Java / Kotlin / Scala
- Python
- Ruby
- Go
- JavaScript / TypeScript
- .NET Framework Based

Patterns & Design Principles

- MVC, MVP, MVVM
- CQRS, Eventual Consistency
- OOP
- Actors
- ACID, CAP Theorem
- SOLID
- TDD
- DDD

Tools

- Git
- GitHub
- Slack
- Trello
- Atlassian Tools

Architecture

- Microservices
- Serverless
- Client / Server
- Layered
- Distributed Systems
- Service Oriented

Security

- Hashing Algorithms
- PKI
- OWASP
- Auth Strategies

Working with Data

- Hadoop, Spark, MapReduce
- ETL, Datawarehouses
- SQL Databases
- NoSQL Databases
- Analytics
  - Apache Spark
  - Hadoop
  - Datawarehouse Principles

Web, Mobile

- Reactive Programming
- Functional Programming
- React, Vue, Angular
- SPA, SSR, SSG
- Microfrontends
- W3C and WHATWG

APIs & Integrations

- gRPC
- ESB, SOAP
- REST
- GraphQL
- BPM, BPEL
- Messaging Queues

Frameworks

- BABOK
- IAF
- UML
- TOGAF

Management

- PMI
- ITIL
- RUP
- Prince2
- Certifications

- LeSS
- SaFE
- XP
- Kanban
- Scrum
- Agile Model

Networks

- OSI
- TCP/IP Model
- HTTP, HTTPS
- Proxies
- Firewalls

Operations Knowledge

- Infrastructure as Code
- Cloud Providers
- Serverless Concepts
- Linux / Unix
- Service Mesh
- CI / CD
- Containers
- Cloud Design Patterns

Visit DevOps Roadmap

Enterprise Software

- MS Dynamics
- SAP ERP, HANA, Business Objects
- EMC DMS
- IBM BPM
- Salesforce

Visit the following relevant roadmaps and keep learning

- Backend
- System Design
- Software Design & Architecture